MILITARY SPECIFICATION

HEATERS, HOT WATER BOOSTER, ELECTRIC; SINK, SANITIZING

This amendment forms a part of Military Specification MIL-H-43895B, dated 24 June 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

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2.1, under "STANDARDS MILITARY": Add the following:

"MIL-STD-167/1 - Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities."

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Add the following new paragraph:

"3.1.1 Human factors criteria (for Navy use only). Human factors engineering criteria, principles and practices, as defined in MIL-STD-1472, shall be incorporated into the design of the hot water booster heaters. The equipment shall be designed such that all maintenance and operational tasks shall permit safe and efficient performance by the 5th percentile female through the 95th percentile male as defined in MIL-STD-1472. Particular attention shall be given to the ease of access for the removal and replacement of the heating elements. All controls, valves, gauges, and indicator lights shall be selected and integrated into the design of the heaters so as to meet the application requirements of MIL-STD-1472 and be clearly and appropriately labeled to identify function. Warning plates, where specified, shall have a white inscription on a red background and shall clearly identify the hazard at hand. The clearance or free area required around an item shall permit an individual with the applicable 5th to 95th percentile body dimensions and physical capabilities as defined in MIL-STD-1472, to safely operate, maintain, remove or replace that item. When establishing accessibility requirements, both physical and visual access must be provided, along with access for any

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tools, test equipment needed and parts replacement. While inspecting for defects and performing tests, the equipment shall adhere to the human factors engineering considerations listed herein."

3.4, line 10, after "long,": Add "8".

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3.4.3.3.1, line 2: Delete "yellow" and substitute "green".

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Add the following new paragraphs:

"3.9 Environmental suitability. When specified for shipboard use (see 6.2), hot water booster heaters shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage when tested as specified in 4.4.2.

3.10 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CE01, CE03, and RE02. (see 4.4.3)."

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4.2.4, line 1, after "4.4.1": Add ", and when required for shipboard use, 4.4.2 and 4.4.3."

line 5, after "4.4.1": Add ", and when applicable, 4.4.2 and 4.4.3".

4.4.1, last line on the page: Delete "yellow" and substitute "green".

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Add the following new paragraphs:

"4.4.2 Shipboard environmental tests. The following tests are required for hot water booster heaters for shipboard use. Failure to operate satisfactorily during or at the conclusion of the tests, or evidence of breakage or excessive looseness of parts shall constitute failure of the test.

a. Inclined test. The hot water booster heater shall be attached to a sink and shall be operated with the sink inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other. The sink shall be filled and heater energized to heat water to a minimum of 180°F.

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The heater shall be operated for 30 minutes in each inclined direction. The temperature shall be monitored to determine conformance to the requirements of 3.4.3.5. The sink and waste collector reservoir shall then be drained. The waste reservoir discharge valve, operating handle, indicator lights and empty reservoir cutout shall be checked for proper operation and conformance to 3.4.2.1, 3.4.3.2.1, and 3.4.3.3.1. Observations shall be made to determine conformance with leak tightness requirements of 3.4. Failure to meet these requirements shall constitute failure of the test.

b. Vibration test. The hot water booster heater shall be attached to a sink in accordance with manufacturer's instructions. The sink shall be filled with water and the heater, under normal operating conditions, shall be tested in accordance with requirements for MIL-STD-167/1, type I equipment. Failure of the heater to perform its function during and after testing shall constitute failure of this test.

4.4.3 Electromagnetic interference tests. When required for shipboard use, the unit shall be tested by the contractor in accordance with test methods CE01, CE03 and RE02 of MIL-STD-462. The contractor shall furnish written certification that the equipment meets MIL-STD-461 requirements. Nonconformance with requirements specified shall constitute failure of this test."

6.2d Delete "and 3.7" and substitute ", 3.7, 3.9 and 3.10".

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 7320-0839

Review activities:

Army - MD, TS
Navy - SA, SH
Air Force - 84
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